IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE PART I: MEng, BEng and ACGI

**Corrected Copy**

**MATHEMATICS 1B (E-STREAM AND I-STREAM)**

Friday, 24 May 10:00 am

Time allowed: 2:00 hours

**There are FOUR questions on this paper.**

**Answer ALL questions. All questions carry equal marks (25% each)**

*NO CALCULATORS ALLOWED.*
*Mathematical Formulae sheet provided*

**Any special instructions for invigilators and information for candidates are on page 1.**

Examiners responsible First Marker(s) : D. Nucinkis
Second Marker(s) : B.K.W. Ee

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# EE1-10B MATHEMATICS II

1. a) (i) Prove the frequency-shift property of the Fourier Transform $\mathscr{F}$, that if $\mathscr{F}[f(t)] = F(\omega)$, then [4]

$$\mathscr{F}\left[e^{iat} f(t)\right] = F(\omega - a).$$

(ii) Show that the Fourier Transform of the function [6]

$$g(t) = \begin{cases} |t| & -1 \le t \le 1 \\ 0 & \text{otherwise} \end{cases},$$

is $\mathscr{F}[g(t)] = 2\text{sinc}\ \omega - \text{sinc}^2(\omega/2)$, where $\text{sinc}\ x = \frac{\sin x}{x}$.

[Hint: consider $e^{-i\omega t} = \cos(\omega t) - i\sin(\omega t)$, and that $|t|$ is even. You may assume $\cos 2x = 1 - 2\sin^2 x$.]

(iii) Hence, or otherwise, compute $\mathscr{F}[g(t)\cos(at)]$. [3]

b) Two planes have equations

$$\Pi_1 : 2x + y + z = 1 \quad \text{and} \quad \Pi_2 : x - 3y + z = 2.$$

(i) Show that $L$, the line of intersection of $\Pi_1$ and $\Pi_2$, has equation

$$\underline{\mathbf{r}} = (-1,0,3) + \lambda(-4,1,7). \qquad [3]$$

(ii) Find $Q$, the point where the plane $\Pi_3 : x + y - z = 6$ intersects the line $L$. [2]

(iii) Obtain the distance between $L$ and the point $P(1,3,-1)$. [4]

(iv) What is the angle at which $\Pi_1$ and $\Pi_2$ meet? [3]


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2. a) Show that for any vectors $\underline{u},\underline{v},\underline{w} \in \mathbb{R}^3$, [ 6 ]

$$[(\underline{u}+\underline{v}) \times (\underline{v}+\underline{w})] \cdot (\underline{w}+\underline{u}) = 2(\underline{u}\times\underline{v}) \cdot \underline{w}$$

b) Consider the matrices

$$A = \begin{pmatrix} 1 & -1 & 1 \\ 3 & -1 & 2 \\ 1 & -2 & 2 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 3 & -2 & 1 \\ -2 & 3 & 1 \\ 1 & -3 & 2 \end{pmatrix}.$$

Obtain (i) $\det(A)$, (ii) $\det(B)$, (iii) $\det(AB)$, and (iv) $\det(BA)$. [ 4 ]

c) You are given a system of linear equations

$$\begin{array}{rcrcrcl} 3x & + & y & - & z & = & \alpha \\ x & - & 3y & + & 2z & = & 5 \\ \beta x & + & 2y & - & 3z & = & 15 \end{array}$$

where $\alpha$ and $\beta$ are constants.

(i) Write the system in the form $A\underline{x} = \underline{b}$ and find $\det(A)$. Hence find the conditions on $\alpha,\beta$ required so that the sytem has (I) infinitely many solutions, (II) no solutions, and (III) a unique solution. In case (I) find all solutions. [ 6 ]

(ii) For $\beta = 8$, find the inverse matrix $A^{-1}$. [ 4 ]

d) Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be a $2\times 2$ matrix. Considering the cases where $b$ or $c$, or both, are zero, or non-zero, find all possible $A$ satisfying [ 5 ]

$$A = A^{-1}.$$

3. a) Given the $2 \times 2$ matrix $A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$,

(i) Find the eigenvalues and eigenvectors for $A$; [ 4 ]

(ii) Hence find an invertible matrix $P$, and a diagonal matrix $D$ satisfying [ 3 ]

$$A = PDP^{-1} \quad \text{and} \quad P^{-1} = P^T;$$

b) Find values of $\alpha$ and $\beta$ so that the first order ODE

$$(2x + \alpha t e^x + \beta \cos t)\frac{dx}{dt} = -e^x + x\sin t + 2t$$

is exact. Hence find the solution satisfying $x(\pi) = 0$. [ 6 ]

c) Solve the Bernoulli equation

$$\frac{dy}{dx} - \frac{2}{x}y = x^2y^4$$

satisfying $y(1) = 1$, by using the substitution $u = \frac{1}{y^3}$. [ 6 ]

d) Find the general solution of the second-order inhomogenous ordinary differential equation [ 6 ]

$$\frac{d^2x}{dt^2} + 4x = 2\sin(2t).$$

4. a) Show that the substitution $x = e^t$ transforms the Euler equation

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + 4y = \frac{1}{x}$$

into the second order linear ODE with constant coefficients

$$\frac{d^2y}{dt^2} + A\frac{dy}{dt} + By = f(t),$$

where $A$, $B$ and $f(t)$ are to be determined.

Hence solve first for $y(t)$ and thus obtain the solution for $y(x)$. [ 7 ]

b) Find the Taylor series approximation of

$$f(x,y) = \ln(1 + x^2 - y)$$

up to quadratic terms, about the point $(0,0)$.

If we choose $x = 0.1$ and $y = -0.1$ to estimate $\ln(1.11)$, we expect this to be correct to how many decimal places? [ 7 ]

c) Obtain all stationary points of the function

$$f(x,y) = 1 - \cos x + \frac{1}{2}y^2,$$

and classify them using the Hessian matrix. [ 6 ]

d) Let $z = f\left(\frac{x}{y}\right)$. Show that [ 5 ]

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = 0.$$


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